Simulation and Modeling Lab -7B1

Question 1: Two different teaching methods are used to teach the same topic. The test scores obtained from students are:

- **Method A:** [72, 68, 75, 71, 74, 69]
- **Method B:** [78, 81, 76, 80, 79, 77]

Assume that the two populations have **equal variances**. At **$\alpha = 0.05$**, investigate whether the two teaching methods produce different average scores.

Tasks:

1. Formulate **H_0 and H_1**.
2. Calculate the **sample mean and sample variance (ddof = 1)** for each group.
3. Find the **difference between the two sample means**.
4. Calculate the **pooled variance**, **t-statistic**, and **degrees of freedom**.
5. Find the **two-tailed p-value**.
6. Construct the **95% confidence interval** for the difference in population means.
7. **Decision & Interpretation:** State your conclusion in **one or two plain-English sentences**.

Question 2: A manufacturer claims that the **average weight of a packaged product is 500 g**. A random sample of 7 packages has the following weights:

- **Data:** [498, 503, 497, 501, 495, 504, 499]
- **Hypothesized mean:** 500 g
- **Tasks:**
    - Calculate the **t-statistic and p-value**.
    - Determine whether the population mean is significantly different from the claimed mean at a **0.05 significance level**.
    - **Interpret and visualize** the result.